	Tutorial set -1 Calculus
1	Let $f(x) = x^3 - x - 1$. **a.** Show that f has a zero between -1 and 2. **b.** Solve the equation $f(x) = 0$ graphically with an error of magnitude at most 10^{-8}. **c.** It can be shown that the exact value of the solution in part (b) is $$\left(\frac{1}{2} + \frac{\sqrt{69}}{18}\right)^{1/3} + \left(\frac{1}{2} - \frac{\sqrt{69}}{18}\right)^{1/3}$$ Evaluate this exact answer and compare it with the value you found in part (b).
2	**Root of an equation** Show that the equation $x + 2\cos x = 0$ has at least one solution.
3	**a.** Graph the function $$f(x) = \begin{cases} x, & -1 \le x < 0 \\ \tan x, & 0 \le x \le \pi/4. \end{cases}$$ **b.** Is f continuous at $x = 0$? **c.** Is f differentiable at $x = 0$? Give reasons for your answers.
4	Find the slope of the curve $x^3y^3 + y^2 = x + y$ at the points $(1, 1)$ and $(1, -1)$.
5	Is the derivative of $$h(x) = \begin{cases} x^2 \sin(1/x), & x \ne 0 \\ 0, & x = 0 \end{cases}$$ continuous at $x = 0$? How about the derivative of $k(x) = xh(x)$? Give reasons for your answers.

6	$\lim_{x\to 1} \frac{x^2+3x-4}{x-1}$ **52.** $\lim_{x\to 1} \frac{x^a-1}{x^b-1}$ $\lim_{x\to \pi} \frac{\tan x}{x}$ **54.** $\lim_{x\to 0} \frac{\tan x}{x+\sin x}$ $\lim_{x\to 0} \frac{\sin^2 x}{\tan(x^2)}$ **56.** $\lim_{x\to 0} \frac{\sin mx}{\sin nx}$
7	An isosceles triangle has its vertex at the origin and its base parallel to the x-axis with the vertices above the axis on the curve $y = 27 - x^2$. Find the largest area the triangle can have.
8	Show that $y = x^2 + \int_1^x \frac{1}{t}\,dt$ solves the initial value problem $\frac{d^2y}{dx^2} = 2 - \frac{1}{x^2}; \quad y'(1) = 3, \quad y(1) = 1.$
9	Find the area of the region cut from the first quadrant by the curve $x^{1/2} + y^{1/2} = a^{1/2}$.
10	Find dy/dx if $y = \int_{\cos x}^{0} (1/(1-t^2))\,dt$. Explain the main steps in your calculation.
11	Use Leibniz's Rule to find the value of x that maximizes the value of the integral $\int_x^{x+3} t(5-t)\,dt.$

12	Let $f(x)$ be a continuous function. Express $$\lim_{n\to\infty} \frac{1}{n}\left[f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\right]$$ as a definite integral. Use the result to evaluate **a.** $\lim_{n\to\infty} \frac{1}{n^2}(2 + 4 + 6 + \cdots + 2n)$, **b.** $\lim_{n\to\infty} \frac{1}{n^{16}}(1^{15} + 2^{15} + 3^{15} + \cdots + n^{15})$,
13	Evaluate $$\lim_{n\to\infty} \frac{1^5 + 2^5 + 3^5 + \cdots + n^5}{n^6}$$ by showing that the limit is $$\int_0^1 x^5\,dx$$ and evaluating the integral.
14	Find the volume of the solid generated by revolving the "triangular" region bounded by the curve $y = 4/x^3$ and the lines $x = 1$ and $y = 1/2$ about **(a)** the x-axis; **(b)** the y-axis; **(c)** the line $x = 2$; **(d)** the line $y = 4$. Find the volume of the solid generated by revolving the region bounded on the left by the parabola $x = y^2 + 1$ and on the right by the line $x = 5$ about **(a)** the x-axis; **(b)** the y-axis; **(c)** the line $x = 5$. Find the volume of the solid generated by revolving the region bounded by the parabola $y^2 = 4x$ and the line $y = x$ about **(a)** the x-axis; **(b)** the y-axis; **(c)** the line $x = 4$; **(d)** the line $y = 4$.

15	Find the lengths of the curves in $y = x^{1/2} - (1/3)x^{3/2}, \quad 1 \le x \le 4$ $x = y^{2/3}, \quad 1 \le y \le 8$ $y = (5/12)x^{6/5} - (5/8)x^{4/5}, \quad 1 \le x \le 32$ $x = (y^3/12) + (1/y), \quad 1 \le y \le 2$
16	Find the centroid of a thin, flat plate covering the region enclosed by the x-axis, the lines $x = 2$ and $x = -2$, and the parabola $y = x^2$. Find the centroid of a thin, flat plate covering the "triangular" region in the first quadrant bounded by the y-axis, the parabola $y = x^2/4$, and the line $y = 4$.
17	find the areas of the surfaces generated by revolving the curves about the given axes. $y = \sqrt{2x + 1}, \quad 0 \le x \le 3; \quad x$-axis $y = x^3/3, \quad 0 \le x \le 1; \quad x$-axis $x = \sqrt{4y - y^2}, \quad 1 \le y \le 2; \quad y$-axis
18	A solid is generated by revolving about the x-axis the region bounded by the graph of the positive continuous function $y = f(x)$, the x-axis, and the fixed line $x = a$ and the variable line $x = b$, $b > a$. Its volume, for all b, is $b^2 - ab$. Find $f(x)$.
19	$\lim_{t\to 0^+} \frac{t - \ln(1 + 2t)}{t^2}$ $\quad$ $\lim_{x\to 4} \frac{\sin^2(\pi x)}{e^{x-4} + 3 - x}$ $\lim_{t\to 0^+} \left(\frac{e^t}{t} - \frac{1}{t}\right)$ $\quad$ $\lim_{y\to 0^+} e^{-1/y} \ln y$ $\lim_{x\to\infty} \left(1 + \frac{3}{x}\right)^x$ $\quad$ $\lim_{x\to 0^+} \left(1 + \frac{3}{x}\right)^x$
20	Find $f'(2)$ if $f(x) = e^{g(x)}$ and $g(x) = \int_2^x \frac{t}{1 + t^4}\, dt$.